Cyber Lab 1

Bationo Jonathan

Description of the experiment

The experiment is about to see how computers communicate over a network through wireshark with the network protocols

Screenshots of wireshark captures

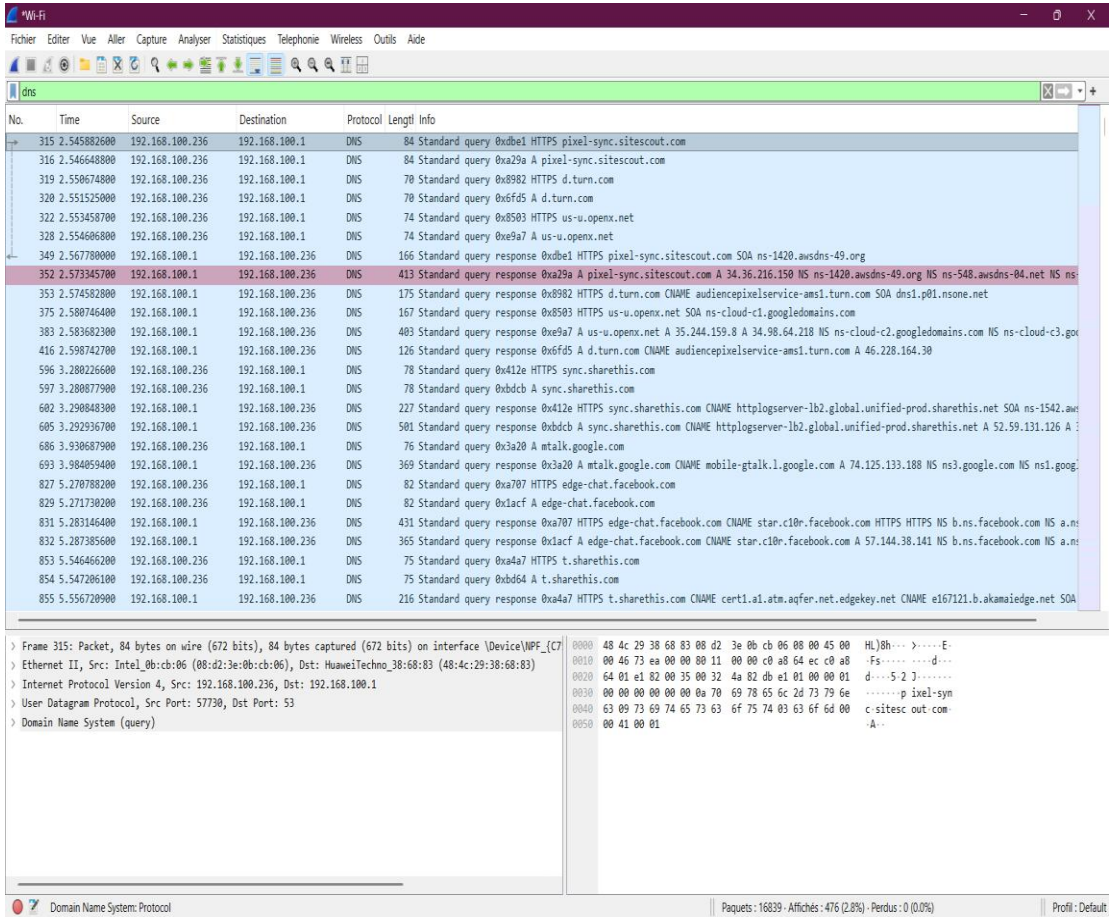

Figure 1

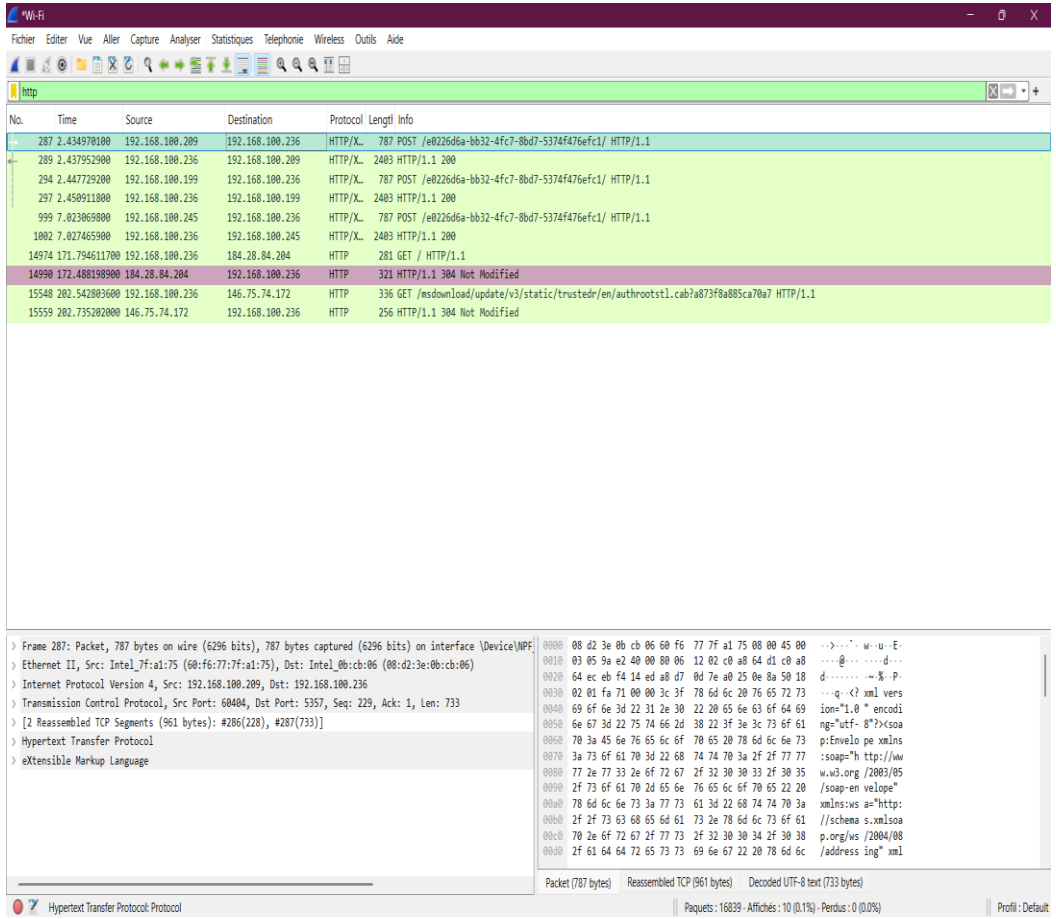

Figure 2

Screenshots of wireshark captures (2)

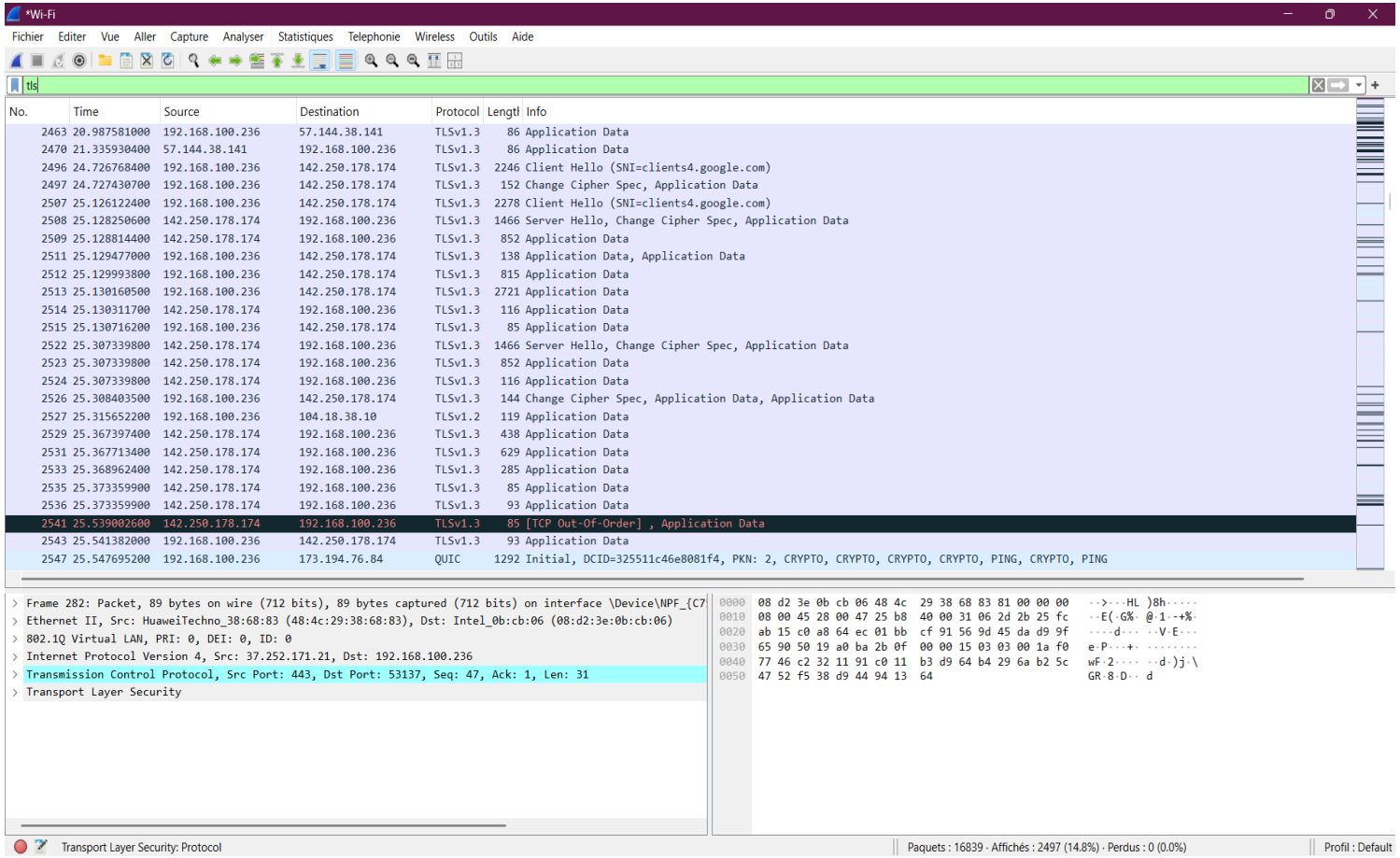

Figure 3

Observed protocols

Tls : The Transport Layer Security (TLS) protocol ensures the confidentiality and integrity of data exchanged on the Internet. (figure 3)

http: HTTP (Hypertext Transfer Protocol) is a fundamental communication protocol for the web, enabling the transfer of information between clients (such as browsers) and servers. (figure 2)

Dns: The Domain Name System (DNS) protocol is an essential Internet system that translates domain names into IP addresses, allowing users to access websites via readable names rather than numbers. (figure 1)

Answering analysis questions

1. **Approximately how many packets were captured?**

   16836 packages

2. **List three protocols observed during the experiment**

   DNS, TLS, HTTP

3. **Why is encryption important when browsing the internet?**

   Encryption is essential when browsing the internet because it secures data in transit, transforming readable information into unreadable code (ciphertext) to prevent unauthorized access, theft, or tampering

4. **What type of information could attackers obtain from unencrypted traffic?**

   Login credentials, financials, personal information

Security lessons learned

We learnt that we have to protect our data by renforce our password with minimum 12 characters, mixture of uppercase and lowercase letters, include numbers, include special characters